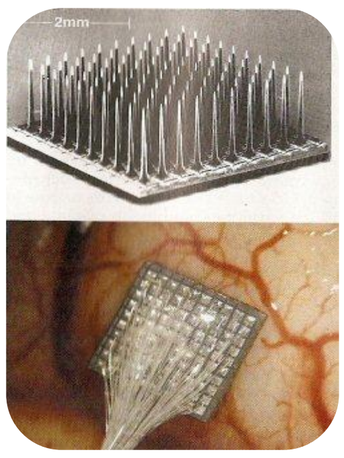
Measuring the Brain to Restore Vision and Motor Function

Marco Ebeling · Predoctoral Research · Universidad Miguel Hernández de Elche

Personalizing Cortical Visual Prostheses for Blind Patients

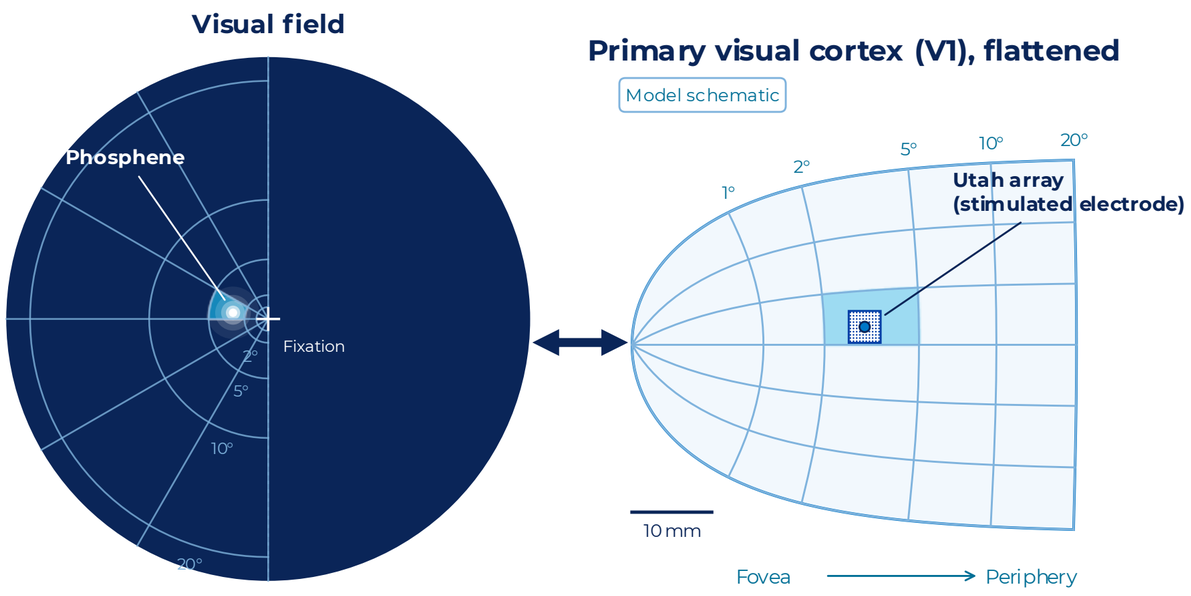
The Clinical Problem

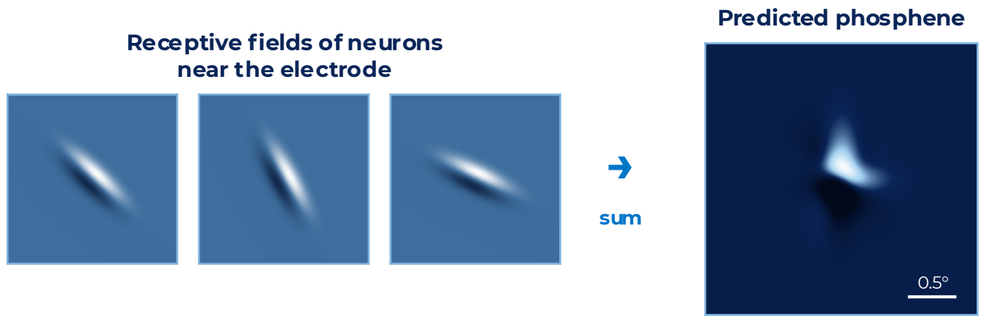
When the eye or optic nerve is destroyed, the visual cortex is the most direct remaining route. The CORTIVIS trial at Universidad Miguel Hernández de Elche implants a 96-electrode array in the primary visual cortex of blind volunteers for six months. The array both stimulates and records neural activity.



From Electrode to Percept

Stimulating an electrode makes the patient see a spot of light, a phosphene. Where it appears depends on how the visual field maps onto V1. Its size and shape depend on the receptive fields of the neurons near the electrode.





The Gap

Models that predict phosphenes have been built mostly in simulation, or with surface electrodes and retinal implants. This clinical trial has access to neural recordings from penetrating intracortical arrays, a missing piece in many prior studies. This project asks how well these models predict what patients see, and whether they can be tailored to each patient.

Next Steps

Test
Run existing phosphene models on stimulation from past patients and compare their predictions with what those patients reported, to find where the models succeed and fail.

Personalize
Build a model tuned to each patient's recordings and reports, so it reflects how their cortex responds instead of relying on typical values. Then check it on past trials it wasn't built from.

Validate
Use the personalized model to choose new stimulation for a patient currently in the clinical trial.

Predicting Motor Recovery after Tumor Removal

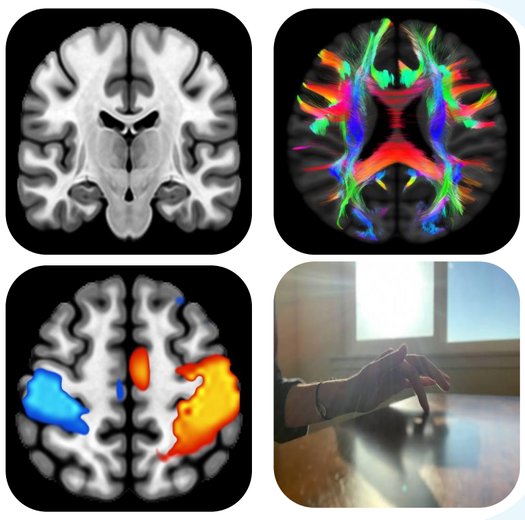
The Clinical Problem

Surgeons removing tumors near the motor cortex or corticospinal tract must take out as much tumor as possible while preserving motor function. However, every tumor is different. Tumors can sit in many places and affect different parts of the motor system. This makes it hard to plan the surgery and to predict who will recover movement afterwards.

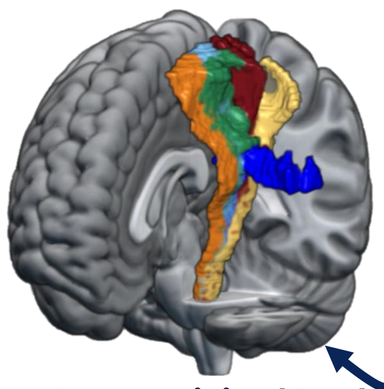
Patients and Scans

Patients operated on at Hospital General Universitario Dr. Balmis in Alicante between 2008 and 2025. Before surgery, each patient had three MRI scans.



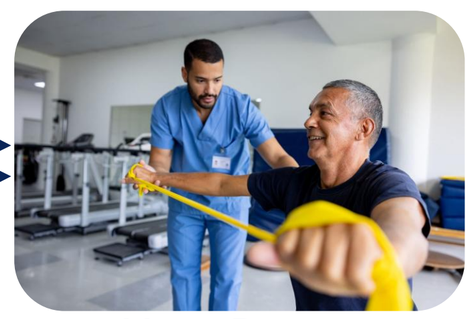


Core Question



DWI shows the integrity of tracts carrying information from the brain to the body. fMRI shows which areas of the brain increase their activity during a task (finger tapping in this study).

Connectivity (DWI)



Motor Recovery

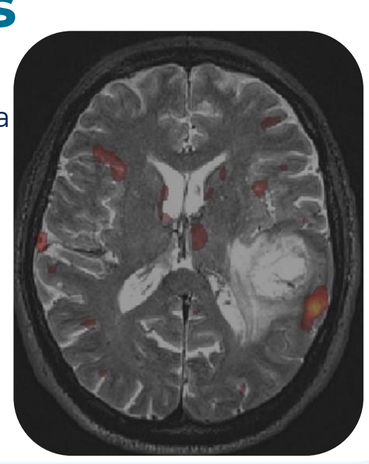


Activity (fMRI)

Each may help predict motor function after surgery, and they may also depend on each other. This project asks how they relate to recovery, whether they work together, and how much.

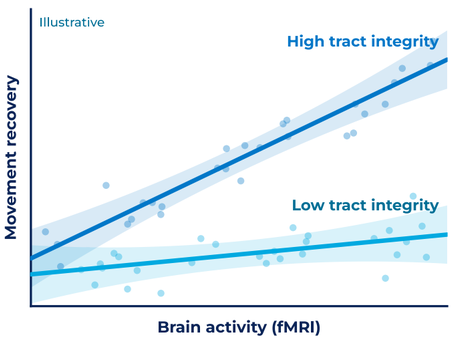
Pathology Challenges

Tumors complicate every step of this process. Swelling around a tumor can distort the scans, and a growing tumor can push brain tissue out of place. There is no standard way to correct for these tumor-specific artifacts. Few studies have combined these modalities, and none has examined how the integrity of individual tract subsegments and motor activation together relate to motor recovery after tumor removal. The first step is to test different analysis methods and find a combination that reliably captures informative metrics.



Next Steps

With reliable measures of connectivity and activity, the next step is to design a study with enough statistical power to test how each modality relates to motor recovery after surgery, and whether they interact. Because the group of subjects is already fixed, a specific set of tests will be defined in advance, while other patterns will be explored more openly. The goal is to improve surgical planning and to identify which patients are most likely to recover before the surgery occurs.



FULBRIGHT España
UNIVERSITAS Miguel Hernández
bio http://nbio.umh.es
marco.ebeling@fulbrightmail.org